

# BIS CRS Exhaustive Master RAG Dataset

Updated Product-by-Product Compliance Breakdown | Statutory Forms | Scheme-II Grouping | Marking & e-Labeling | Regulatory Circulars | Standard Specifics & Clauses

## 1. Product-by-Product Compliance Breakdown (All 15 IT Categories)

### 1. IT Power Adapters & External Power Supplies

|                                       |                                                                                                                                                                                       |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Applicable IS Standards               | IS 13252 (Part 1) / IS/IEC 62368-1: 2018/2023                                                                                                                                         |
| Scope & Technical Inclusions          | AC-DC power supply adapters for laptops, routers, desktop computers, PoE injectors, external SSD/HDDs, wearables, and extended reality (XR) headsets.                                 |
| Exclusions                            | Industrial high-voltage power units (>600V) and open-frame embedded industrial supplies.                                                                                              |
| Key Safety & Test Parameters          | Dielectric strength, high-voltage leakage current testing, insulation breakdown, creepage and clearance distances (>=5mm), touch temperature limits, and short-circuit fault testing. |
| Safety Critical Components List (CCL) | Optocouplers, isolation transformers, Y-capacitors, MOV varistors, internal batteries, PCB flame ratings (UL94 V-0), and thermistors.                                                 |

### 2. Laptops & Notebooks

|                                       |                                                                                                                                                                            |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Applicable IS Standards               | IS 13252 (Part 1) / IS/IEC 62368-1: 2018/2023                                                                                                                              |
| Scope & Technical Inclusions          | Portable computers, ultrabooks, touch notebooks, mini-portables, and convertible laptops with integrated displays.                                                         |
| Exclusions                            | Industrial rackmount servers and ruggedized custom military computing units.                                                                                               |
| Key Safety & Test Parameters          | Enclosure drop test (1 meter), battery charging management safety, thermal runaway protection, keyboard touch temperature limits (<48 C), and fire enclosure flammability. |
| Safety Critical Components List (CCL) | Lithium battery pack, internal power adapter, motherboard PCB (UL94 V-0), cooling fan assembly, and internal wiring.                                                       |

### 3. Mobile Phones & Smartphones

|                                       |                                                                                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Applicable IS Standards               | IS 13252 (Part 1) / IS/IEC 62368-1: 2018/2023                                                                                                               |
| Scope & Technical Inclusions          | Smartphones, feature phones, satellite handhelds, and cellular IoT handheld voice/data terminals.                                                           |
| Exclusions                            | Fixed wireless desktop phones without cellular mobility and non-cellular walkie-talkies.                                                                    |
| Key Safety & Test Parameters          | Specific Absorption Rate (SAR) compliance, battery overcharge protection, secondary cell isolation, short-circuit fault testing, and acoustic shock limits. |
| Safety Critical Components List (CCL) | Secondary Lithium-ion cell (certified to IS 16046 Part 2), battery protection IC (BMS), charger adapter, and display panel assembly.                        |

### 4. Self-Ballasted LED Lamps

|                                       |                                                                                                                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Applicable IS Standards               | IS 16102 (Part 1) / IS 16102 (Part 2)                                                                                                                                           |
| Scope & Technical Inclusions          | Mains-voltage retrofittable LED bulbs for domestic and general lighting applications (E27, B22, E14, B15d caps).                                                                |
| Exclusions                            | Low-voltage DC LED strip lights and automotive headlight LED modules.                                                                                                           |
| Key Safety & Test Parameters          | Insulation resistance, mechanical cap torque strength (>3Nm), thermal endurance, creepage/clearance distance verification, photobiological safety, and harmonic current limits. |
| Safety Critical Components List (CCL) | LED driver module, flame-retardant housing, internal fusible resistor, lamp cap contact assembly, and integrated rechargeable battery (if applicable).                          |

### 5. LED Control Gear / Drivers

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| Applicable IS Standards | IS 15885 (Part 1) / IS 15885 (Part 2 / Section 13) |
|-------------------------|----------------------------------------------------|

|                                              |                                                                                                                                                                                                                                     |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope &amp; Technical Inclusions</b>      | AC-DC or DC-DC independent and built-in LED drivers/lamp controlgear for luminaires used on DC supplies up to 250 V and/or AC supplies up to 1 100 V at 50 Hz.                                                                      |
| <b>Exclusions</b>                            | Resistor-based passive current limiters and non-LED power transformers.                                                                                                                                                             |
| <b>Key Safety &amp; Test Parameters</b>      | Type testing, thermal overload cut-off, output short-circuit endurance, isolation transformer insulation class, thermally protected controlgear checks (Annex B), overheating protection (Annex C), and high-voltage flash testing. |
| <b>Safety Critical Components List (CCL)</b> | Isolation transformer, electrolytic capacitors, internal fuse, thermal cut-offs, and enclosure polymer material.                                                                                                                    |

## 6. Sealed Secondary Lithium Batteries & Cells

|                                              |                                                                                                                                                                                                                  |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Applicable IS Standards</b>               | IS 16046 (Part 2) / IEC 62133-2: 2017                                                                                                                                                                            |
| <b>Scope &amp; Technical Inclusions</b>      | Sealed secondary Lithium-ion and Lithium-polymer cells, coin cells, and multi-cell battery packs used in portable electronics.                                                                                   |
| <b>Exclusions</b>                            | Primary non-rechargeable lithium coin cells, nickel systems (covered in Part 1), and lead-acid accumulator batteries.                                                                                            |
| <b>Key Safety &amp; Test Parameters</b>      | Continuous charging under constant voltage, forced internal short-circuit, mechanical shock, vibration (Clause 7.3.8.1/7.3.8.2), 130 C thermal abuse testing, and assembly of cells into batteries (Clause 5.6). |
| <b>Safety Critical Components List (CCL)</b> | Cell separator film, Battery Management System (BMS) / protection circuit board, PTC thermistor, and outer protective sleeve.                                                                                    |

## 7. Sealed Secondary Nickel Batteries & Cells

|                                              |                                                                                                                                   |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Applicable IS Standards</b>               | IS 16046 (Part 1)                                                                                                                 |
| <b>Scope &amp; Technical Inclusions</b>      | Sealed secondary Nickel-Cadmium (Ni-Cd) and Nickel-Metal Hydride (Ni-MH) rechargeable cells and battery packs.                    |
| <b>Exclusions</b>                            | Open vented wet nickel cells and industrial non-sealed storage accumulators.                                                      |
| <b>Key Safety &amp; Test Parameters</b>      | Continuous low-rate charging endurance, deep discharge safety, pressure relief vent operational check, and mechanical drop tests. |
| <b>Safety Critical Components List (CCL)</b> | Safety vent valve, cell metallic can, positive/negative plate substrates, and battery protection circuit.                         |

## 8. Smart Watches & Fitness Trackers

|                                              |                                                                                                                                                                           |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Applicable IS Standards</b>               | IS 13252 (Part 1) / IS/IEC 62368-1: 2018/2023                                                                                                                             |
| <b>Scope &amp; Technical Inclusions</b>      | Wearable activity trackers, smartwatches with Bluetooth, Wi-Fi, or Cellular connectivity, and biometric bands.                                                            |
| <b>Exclusions</b>                            | Passive non-electronic wrist straps and medical-grade implantable monitoring devices.                                                                                     |
| <b>Key Safety &amp; Test Parameters</b>      | Ingress protection against sweat corrosion, lithium battery protection circuit (BMS) integrity, enclosure skin-contact surface temperature safety, and drop impact tests. |
| <b>Safety Critical Components List (CCL)</b> | Rechargeable Li-Po cell, magnetic charging puck/cable, mainboard PCB, and bio-compatible enclosure polymer.                                                               |

## 9. Wireless Keyboards & Mice

|                                              |                                                                                                                                       |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Applicable IS Standards</b>               | IS 13252 (Part 1) / IS/IEC 62368-1: 2018/2023                                                                                         |
| <b>Scope &amp; Technical Inclusions</b>      | 2.4GHz RF and Bluetooth desktop entry accessories, wireless optical mice, trackballs, and touchpads.                                  |
| <b>Exclusions</b>                            | Wired USB keyboards/mice powered exclusively by host bus power without internal RF transceivers or batteries.                         |
| <b>Key Safety &amp; Test Parameters</b>      | Low-voltage power supply safety, internal battery charging circuit safety, RF module power density limits, and mechanical drop tests. |
| <b>Safety Critical Components List (CCL)</b> | Main PCB (UL94 V-0 flame rating), internal rechargeable cell (if applicable), and RF transceiver module.                              |

## 10. Set Top Boxes

|                                              |                                                                                                                                            |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Applicable IS Standards</b>               | IS 13252 (Part 1) / IS/IEC 62368-1: 2018/2023                                                                                              |
| <b>Scope &amp; Technical Inclusions</b>      | DTH satellite receivers, digital cable TV set-top boxes, and IPTV/OTT streaming media players.                                             |
| <b>Exclusions</b>                            | Commercial broadcasting station transmitters and head-end video processors.                                                                |
| <b>Key Safety &amp; Test Parameters</b>      | Power supply insulation breakdown, continuous operation temperature rise, and needle-flame fire retardation testing of the plastic casing. |
| <b>Safety Critical Components List (CCL)</b> | External power adapter or internal PSU, main system SoC, and flame-retardant outer plastic enclosure.                                      |

## 11. Automatic Data Processing Machines (Desktop Computers)

|                                              |                                                                                                                                                 |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Applicable IS Standards</b>               | IS 13252 (Part 1) / IS/IEC 62368-1: 2018/2023                                                                                                   |
| <b>Scope &amp; Technical Inclusions</b>      | All-in-One PCs, desktop workstations, and small-form-factor tower personal computers.                                                           |
| <b>Exclusions</b>                            | Industrial Programmable Logic Controllers (PLCs), mainframes, and supercomputers.                                                               |
| <b>Key Safety &amp; Test Parameters</b>      | Grounding continuity (<0.1 ohm), touch current leakage limits (<3.5mA), internal power supply (SMPS) safety, and earth continuity verification. |
| <b>Safety Critical Components List (CCL)</b> | Internal SMPS unit, motherboard, CPU cooling assembly, and metal enclosure chassis grounding.                                                   |

## 12. Power Banks

|                                              |                                                                                                                                                         |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Applicable IS Standards</b>               | IS 13252 (Part 1) & IS 16046 (Part 2)                                                                                                                   |
| <b>Scope &amp; Technical Inclusions</b>      | Portable energy storage banks used for charging mobile devices via USB output ports.                                                                    |
| <b>Exclusions</b>                            | Industrial solar energy storage generators (>500Wh) and vehicle jump-starter lead-acid packs.                                                           |
| <b>Key Safety &amp; Test Parameters</b>      | Dual protection IC for overcurrent/overvoltage, 1-meter drop test, cell balancing control, thermal dissipation limits, and short-circuit fault testing. |
| <b>Safety Critical Components List (CCL)</b> | Internal Lithium cells, main protection board (BMS), charging controller IC, and flame-retardant casing material.                                       |

## 13. CCTV Cameras & Recorders

|                                              |                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Applicable IS Standards</b>               | IS 13252 (Part 1) / IS 16864                                                                                                                      |
| <b>Scope &amp; Technical Inclusions</b>      | Network IP security cameras, dome/bullet analog cameras, Network Video Recorders (NVR), and Digital Video Recorders (DVR).                        |
| <b>Exclusions</b>                            | Military thermal imaging scopes and body-worn law enforcement cameras.                                                                            |
| <b>Key Safety &amp; Test Parameters</b>      | Power over Ethernet (PoE) dielectric isolation, image sensor heat dissipation, external power supply certification, and surge voltage resistance. |
| <b>Safety Critical Components List (CCL)</b> | Internal PSU board, image processor board, PoE module, and outer housing assembly.                                                                |

## 14. Smart Speakers & Audio Amplifiers

|                                              |                                                                                                                                       |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Applicable IS Standards</b>               | IS 616 / IS/IEC 62368-1: 2018/2023                                                                                                    |
| <b>Scope &amp; Technical Inclusions</b>      | Powered active loudspeakers, soundbars, smart voice assistants, audio/video apparatus, and home theater audio amplifiers.             |
| <b>Exclusions</b>                            | Passive non-amplified speaker drivers and industrial public address system arrays.                                                    |
| <b>Key Safety &amp; Test Parameters</b>      | Acoustic impulse pressure limits, transformer thermal fuse verification, and amplifier breakdown tests under single-fault conditions. |
| <b>Safety Critical Components List (CCL)</b> | Mains power transformer, power amplifier PCB, thermal cut-off switches, and acoustic enclosure casing.                                |

## 15. Uninterruptible Power Systems (UPS) & Inverters

|                                |                                                                         |
|--------------------------------|-------------------------------------------------------------------------|
| <b>Applicable IS Standards</b> | IS 16242 (Part 1: 2025) / IEC 62040-1: 2017 + AMD1:2021 + AMD2:2022 CSV |
|--------------------------------|-------------------------------------------------------------------------|

|                                              |                                                                                                                                                                              |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope &amp; Technical Inclusions</b>      | Offline, line-interactive, and online UPS systems up to 2kVA output capacity for office and IT equipment.                                                                    |
| <b>Exclusions</b>                            | Industrial three-phase online UPS systems (>10kVA) and grid-tied solar central inverters.                                                                                    |
| <b>Key Safety &amp; Test Parameters</b>      | Inverter transfer time, output short-circuit clearance time, battery reverse polarity protection, safety requirements alignment, and transfer switch isolation gap distance. |
| <b>Safety Critical Components List (CCL)</b> | Power inverter board, mechanical transfer relays, battery connecting leads, internal battery pack, and circuit breaker.                                                      |

## 2. Standard Forewords, Scope Clauses & Amendments (Specific Standard Technical Mapping)

### IS 16102 (Part 1) : 2026

**Title:** Self-Ballasted LED Lamps for General Lighting Services - Part 1 Safety Requirements

ICS: 29.140.99 | Committee: ETD 23

**Scope & Foreword:** Specifies safety and interchangeability requirements, test methods, and compliance conditions for self-ballasted LED lamps intended for domestic and similar general lighting purposes. Covers rated power up to 60 W; voltage above 50 V AC up to 250 V AC at 50 Hz and DC up to 1000 V; Edison screw caps (E14, E27) and Bayonet caps (B22d, B15d). Includes photobiological safety, rechargeable batteries (Annex B), and smart features (communication/sensors/speakers/charging).

**Reference Standards / Clauses:** IS 8913:2025 / IEC 60360:1998 (Lamp cap temp rise); IS 9206:1979 (Dimensions of caps for tungsten lamps); IS 10322 (Part 1):2014 (Luminaires: General requirements); IS 1401:2008 / IEC 61032:1997 (Probes for verification); IS 15518 (Part 1):2004 (Safety for incandescent lamps); IS 15885 (Part 1):2011 (Safety of lamp controlgear); IS 16108:2012 / IEC 62471:2006 (Photobiological safety); IS 16661:2019 / IEC/TR 62778:2014 (Blue light hazard assessment); IS/IEC 60529:2001 (IP Code enclosure protection); IS/IEC 60695 Series (-2-10, -2-11, -2-12, -2-13: 2021) (Glow-wire fire hazard test methods); ISO 4046-4:2016 (Paper/board vocabulary); IEC 60061-3 (Lamp caps, holders, gauges); IEC 61347-1:2015/AMD1:2017 (Lamp controlgear).

### IS 15885 (Part 1) : 2011

**Title:** Safety of Lamp Controlgear - Part 1 General Requirements

ICS: 29.140.99 | Committee: ETD 23 | Status: Reaffirmed 2021

**Scope & Foreword:** General and safety requirements for lamp controlgear for use on DC supplies up to 250 V and/or AC supplies up to 1 100 V at 50 Hz. Specified as type tests.

**Reference Standards / Clauses:** Annex B: General and safety requirements applicable to thermally protected lamp controlgear. Annex C: Additional general and safety requirements for electronic lamp controlgear with protection against overheating.

### IS 16046 (Part 2) : 2018 / IEC 62133-2 : 2017

**Title:** Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes - Safety Requirements for Portable Sealed Secondary Cells and for Batteries Made from Them for Use in Portable Applications - Part 2 Lithium Systems

ICS: 29.220.30 | Committee: ETD 11

**Scope & Foreword:** Identical with IEC 62133-2 : 2017. Second revision separating nickel systems into Part 1, incorporating coin cell requirements, cell assembly into batteries (Clause 5.6), mechanical tests [vibration, shock] (Clause 7.3.8.1, 7.3.8.2), and IEC TR 62914 reference.

### IS 13252 (Part 1) : 2010 / IEC 60950-1 : 2005

**Title:** Information Technology Equipment - Safety - Part 1 General Requirements

ICS: 33.020; 35.260 | Committee: LITD 16 | Status: Reaffirmed 2022

**Scope & Foreword:** Identical with IEC 60950-1 : 2005 (Second Revision). Provides general safety requirements for IT equipment. Standard practice uses point (.) as decimal marker in place of comma (,).

### IS 16242 (Part 1) : 2025 / IEC 62040-1 : 2017 + AMD1:2021 + AMD2:2022 CSV

**Title:** Uninterruptible Power Systems (UPS) - Part 1 Safety Requirements

ICS: 29.200 | Committee: ETD 31

**Scope & Foreword:** Identical with IEC 62040-1:2017 + AMD1:2021 + AMD2:2022 CSV. First revision updating safety requirements for UPS systems to align with international practices.

### **IS/IEC 62368 (Part 1) : 2023**

**Title:** Audio/Video, Information and Communication Technology Equipment - Part 1 Safety Requirements

ICS: 33.160.01; 35.020 | Committee: LITD 07

**Scope & Foreword:** Identical to IEC 62368-1:2023 (Second Revision). Covers audio/video, IT, and communication technology equipment safety.

### **IS 616 : 2017**

**Title:** Audio, Video and Similar Electronic Apparatus - Safety Requirements (Fifth Revision)

Price Ref: 1,730.00

**Scope & Foreword:** Safety requirements for audio, video, and electronic apparatus.

## **3. BIS CRS Statutory Forms & Declarations**

**Form I** - Statutory Application for Registration under Scheme-II: Collects essential applicant details including brand trademark, physical factory address, and target model list. Requires signature of CEO or designated legal signatory.

**Form II** - Legal Undertaking & Affidavit: Legally binding undertaking confirming adherence to statutory compliance, market surveillance sample collection, and penal liabilities under BIS Act 2016. Notarized on non-judicial stamp paper.

**Form III** - Authorized Indian Representative (AIR) Designation Form: Mandatory for overseas manufacturers without an Indian office. AIR must possess valid GST, IEC, and PAN credentials.

**Form IV** - Laboratory Test Report Ingestion Summary & CCL Declaration: Integrates NABL-accredited laboratory test findings with Safety Critical Components List (CCL). Holds a strict 90-day validity window from issuance.

## **4. Series Grouping & Technical Rules (Scheme-II)**

**Maximum Grouping Capacity:** Maximum 10 models per registration number sharing identical design/application features.

**Manufacturing Location:** Must be manufactured at the exact same physical factory address.

**Brand Uniformity:** All models must bear an identical brand name/trademark.

**Base Model Representative Testing:** Model with highest power rating/wattage/complexity is tested as Base Model. Secondary models cannot exceed the base model's electrical envelope.

**Constructional & Layout Constraints:** Grouped models must feature identical PCB insulation layouts, transformer classes, and enclosure flammability ratings (UL94 V-0 or V-1).

## **5. Marking, Standard Mark & e-Labelling Guidelines**

**Standard Mark Display Format:** Indian Standard number on top, official BIS self-declaration logo in middle, Registration Number (R-XXXXXXX) at bottom.

**Portal URL Requirement:** [www.bis.gov.in](http://www.bis.gov.in) must be printed directly below the Registration Number.

**Dimensional Specifications:** BIS logo symbol minimum height: 6mm; text/numbers minimum height: 1mm.

**e-Labelling Provisions:** Permitted for devices with screens (smartphones, laptops, smartwatches). Standard mark must be accessible within max 3 navigation steps (Settings -> About -> Regulatory).

## **6. Regulatory Circulars & Laboratory Subcontracting (Circular CMD-1/2:4:1)**

**Recommendatory Levels of Control:** 'Levels of Control' given in Scheme of Inspection and Testing (SIT) are recommendatory. Manufacturers can define custom batch/control units with BO notification.

**Subcontracting 'R' (In-House) Category Tests:** MSME manufacturers under Scheme-I can subcontract 'R' category tests to BIS-recognized, BIS-empanelled, or NABL-accredited laboratories under IS/ISO/IEC 17025.

**Subcontracting 'S' (Out-House) Category Tests:** Can be subcontracted to recognized/empanelled or NABL-accredited labs under IS/ISO/IEC 17025.

Record Keeping & Risk-Based Surveillance: Complete records of external tests must be maintained. BIS dynamically reviews surveillance frequency based on risk assessment.

Scope Exceptions: Does not apply to DPIIT/PESO regulated products under pre-dispatch inspection, cement, food products, or manuals issued on/after 15 Jan 2024. Supersedes circular dated 19 Jan 2024.

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